SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: **Artificial Intelligence** **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO2: *Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)*

Assessment Tool Weightage: 40%

QUESTIONS: 3

Duration: 1 Hour
Total Marks:30

Annexure A

Dry Run Evaluation

Code of Conduct:

I SK.Arshad Basha (192312336) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

SK.Arshad Basha

20+28+20+18+8

= 94

...ithmetic CSP

...D + MORE = MONEY. Each letter represents a unique digit ...-9 and m cannot be 0. perform constraint propagation, find a valid assignment & verify the addition.

A) **Problem**

SEND + MORE = MONEY

Each letter represent a unique digit from 0-9 & $m \neq 0$

```
  SEND
+ MORE
------
 MONEY
```

Introduce carried

c_1, c_2, c_3, c_4

From right to left

$D + E = Y + 10c_1$

$N + R + c_1 = E + 10c_2$

$E + O + c_2 = N + 10c_3$

$S + M + c_3 = O + 10c_4$

$c_4 = m = 1$

constraint propagation gives

$S = 9,\quad E = 5\quad N = 6\quad D = 7$

$m = 1\quad O = 0\quad R = 8\quad Y = 2$

Valid Assignment

Letter	S	E	N	D	M	O	R	Y
Digit	9	5	6	7	1	0	8	2

∴ SEND = 9567

MORE = 1085

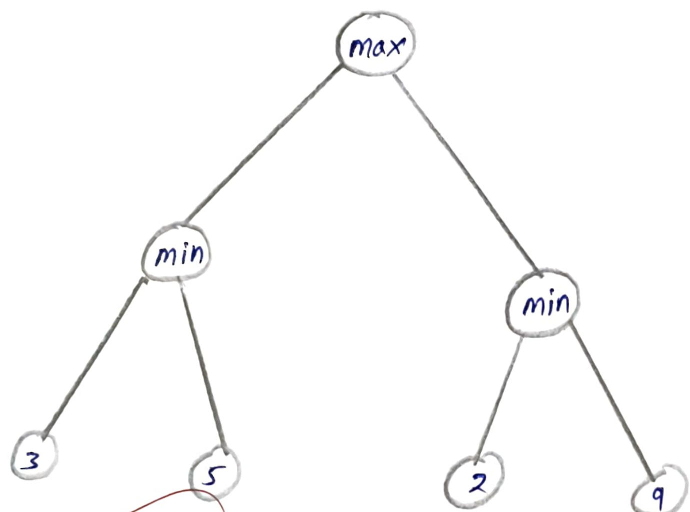
MONEY = 10652

<u>verification</u>

9567 + 1085 = 10652

Hence, the cryptarithm is satified.

2) performe the minimax algorithm on displayed game tree. calculated the value of each min node, the max root value & the move seleeted by max.



Calculate MIN Nodes

Left min :-

min (3, 5) = 3

Right min :-

min (2, 9) = 2

Calculate max root

max (3, 2) = 3

Node	Children values	value
1 - max	3, 2	3
2 - min	3, 5	3
3 - min	2, 9	2
4 - Leaf	-	3
5 - Leaf	-	5

max root value = 3

optimal move :- max choose the left min node, because

$3 > 2$

3) consider a max root with two min children. Each min child has two max children. The leaf vawle from left to right 3,5 | 2,9 | 4,6 | 1,7 perform a complete minimax dry run & identify the optimal value

A) Given leaf values from left to right

3,5 | 2,9 | 4,6 | 1,7

step 1:- max nodes

Node 4:-

max (3,5) = 5

Node 5 :-

max (2,9) = 9

Node 6 :-

max (4,6) = 6

Node 7

max(1,7) = 7

Step 2: min nodes

Node 2

min(5,9) = 5

Node 3 :-

min(6,7) = 6

Step 3:- max root

max(5,6) = 6

Table

Node	children value	value
1 - max	5, 6	6
2 - min	5, 9	5
3 - min	6, 7	6
4 - max	3, 5	5
5 - max	2, 9	9
6 - max	4, 6	6
7 - max	1, 7	7

max root value = 6

The optimal move is the right min child (Node 3) because

max(5,6) = 6

[illegible] move = Right subtree